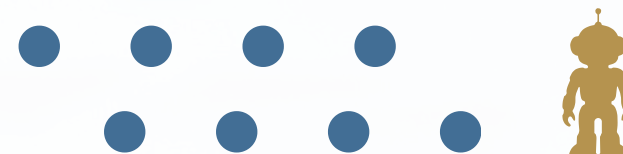




# Enhancing Credit Card Fraud Detection

Using Machine Learning

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# Current challenge

Every missed fraud leads to financial losses and damages customer trust.

Every false fraud alert interrupts legitimate customer transactions.

## Our objective

Find the machine learning model that maximizes fraud detection while minimizing unnecessary customer friction.

Security



Customer  
Experience





# Research workflow



## Dataset

- 1M credit card transactions
- 7 fraud-related variables
- No missing values
- Highly imbalanced



## Research strategy

- Tested 7 different machine-learning models
- Applied 80/20 train-test split
- Verified correlation between variables and target
- Compared preprocessing techniques (normalization and oversampling)



## Evaluation criteria

- Accuracy, Precision, Recall and ROC-AUC
- Confusion matrices to analyze false positives and false negatives
- 5-fold cross-validation to assess model stability







Threshold optimization for Random Forest improved fraud detection, enabling zero missed fraud cases with a small increase in customer verification.

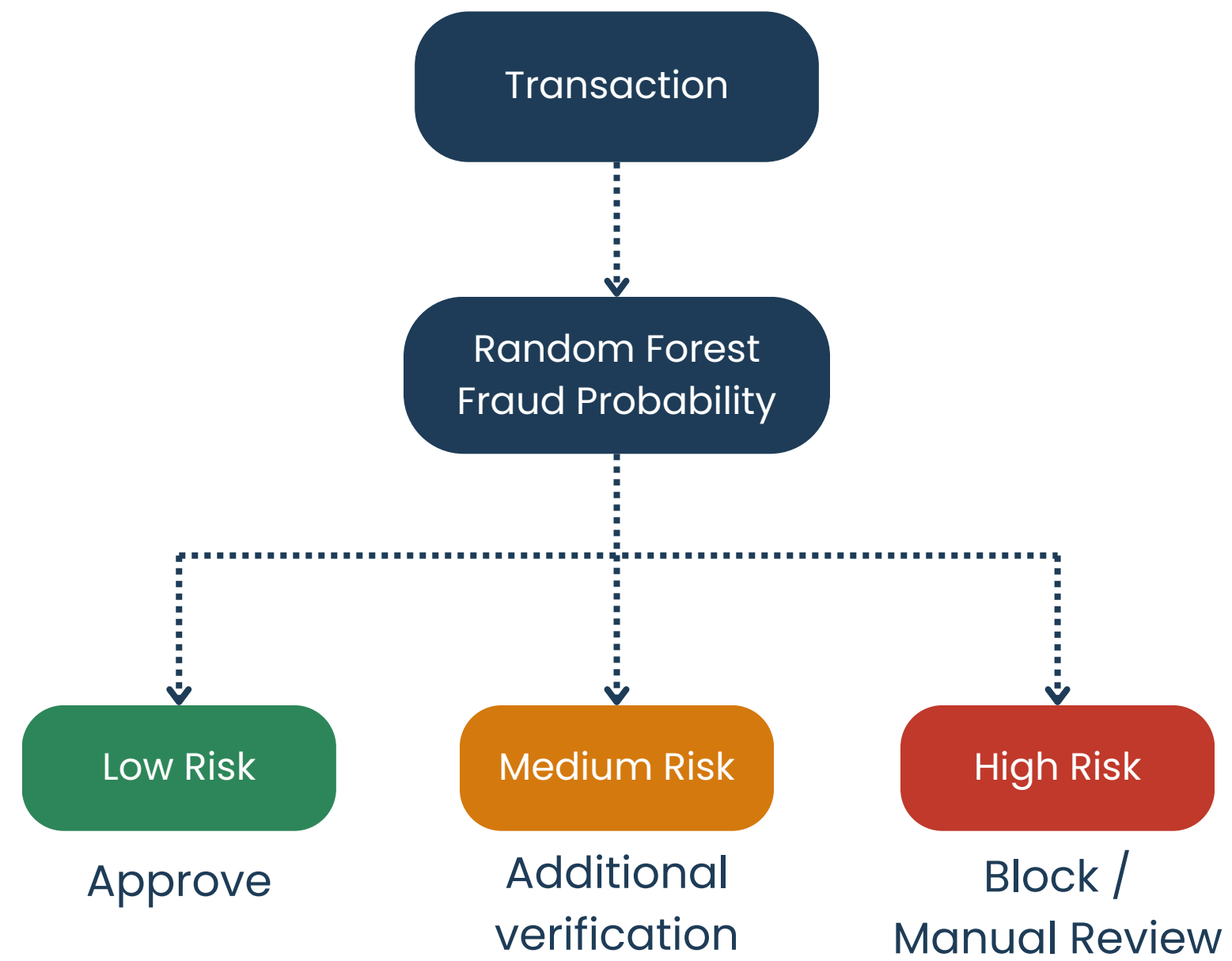


# Key Findings

| Model         | Precision | Recall | ROC-AUC |
|---------------|-----------|--------|---------|
| Random Forest | 1         | 0.999  | 1       |
| Decision Tree | 0.999     | 0.999  | 0.999   |
| Bagging       | 0.999     | 0.999  | 0.999   |

| Decision Threshold | False Negatives | False Positives |
|--------------------|-----------------|-----------------|
| 0.50 (Default)     | 2               | 0               |
| 0.4                | 1               | 0               |
| 0.1                | 0               | 26              |





# Deployment strategy

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# Summary

## Recommendation:

Deploy the Random Forest model as the bank's fraud detection engine.

## Why?

- Highest overall predictive performance
- Excellent cross-validation stability
- Flexible decision threshold
- Supports risk-based authentication
- Reduces missed fraud while maintaining customer experience







# Thank you!

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